1.2 Atoms, Molecules & Stoichiometry

Question Paper

Course	CIE A Level Chemistry
Section	1. Physical Chemistry
Topic	1.2 Atoms, Molecules & Stoichiometry
Difficulty	Easy

Time allowed: 80

Score: /66

Percentage: /100

Question 1a

Explain what is meant by the terms

Relative atomic mass

[2]

Relative isotopic mass

[2]

[4 marks]

Question 1b

i)

State the symbols used for relative atomic mass and relative molecular mass.

[1]

ii)

Describe how to calculate the relative molecular mass of a compound.

[1]

iii)

State the difference between relative molecular mass and relative formula mass.

[1]

[3 marks]

Question 1c

Ionic compounds such as sodium chloride are arranged in a giant three-dimensional lattice structure, with a regular and repeating pattern of positive and negative ions.

State the formula unit of sodium chloride.

[1 mark]

Question 1d

Complete Table 1.1

Table 1.1

Chemical	Formula	Relative molecular / formula mass
Magnesium oxide	MgO	
Sulfuric acid	H_2SO_4	
Ammonium phosphate	$(NH_4)_3PO_4$	
Copper(II) sulfate pentahydrate	$CuSO_4{\bullet}5H_2O$	

[4 marks]

Question 2a

Give the number of particles in one mole of a chemical.

[1 mark]

Question 2b

i) Give the number and type of atoms present in a molecule of water.

[2]

ii)

Calculate the total number of particles and atoms present in one mole of water.

[1]

[3 marks]

Question 2c

i)

Give the number and type of ions present in one mole of sodium carbonate, Na_2CO_3.

[2]

ii)

Calculate the number of metal ions present in **two** moles of sodium carbonate.

[2]

[4 marks]

Question 2d

i) State the conditions when one mole of a gas occupies a volume of 24.0 dm^3.

[2]

ii)

Complete Table 2.1.

Table 2.1

Gas	Number of moles	Volume of gas (dm^3)	Number of molecules present
Nitrogen	2.0	48.0	
Sulfur dioxide		1.8	
Carbon monoxide			9.03×10^{23}

[5]

[7 marks]

Question 3a

Complete Table 3.1.

Table 3.1

Ion	Formula
Nitrate	NO_3^-
	CO_3^{2-}
Sulfate	
	OH^-
	NH_4^+
Zinc(II)	
	Ag^+
	HCO_3^-
Phosphate	

[3 marks]

Question 3b

Complete Table 3.2.

Table 3.2

Ionic compound	Formula
Sodium chloride	NaCl
Potassium oxide	
	$Al_2(CO_3)_3$
Copper(II) nitrate	

[3 marks]

Question 3c

Aqueous silver nitrate should not be acidified with hydrochloric acid as it forms a white precipitate of silver chloride, AgCl, and nitric acid.

i)
Write the chemical equation for the reaction of silver nitrate with hydrochloric acid. Your answer should include state symbols.

[2]

ii)
Silver ions from the silver nitrate and hydrogen ions from the hydrochloric acid are responsible for the formation of silver chloride.

$$Ag^+ + Cl^- \rightarrow AgCl$$

Suggest why the nitrate ions and hydrogens do not appear in this ionic equation.

[1]

[3 marks]

Question 3d

Aqueous barium chloride should not be acidified with sulfuric acid as it forms a white precipitate of barium sulfate and hydrochloric acid.

i)
Write the chemical equation for the reaction of barium chloride with sulfuric acid. Your answer should include state symbols.

[2]

ii)
Write the ionic equation for the formation of barium sulfate.

[1]

[3 marks]

Question 4a

Copper(II) chloride can be produced by the reaction of hydrochloric acid with different copper(II) compounds.

i)

Suggest why the reaction of copper with hydrochloric acid is not used to form copper(II) chloride.

[1]

ii)

Complete Table 4.1 by writing the chemical equation for the formation of copper(II) chloride using hydrochloric acid and the copper(II) compounds.

Table 4.1

Copper(II) compound	Chemical equation
Copper(II) oxide, CuO	....
Copper(II) hydroxide, $Cu(OH)_2$	....
Copper(II) carbonate, $CuCO_3$	....

[3]

[4 marks]

Question 4b

11.93 g of copper(II) oxide is reacted with 50 cm^3 of 2.0 mol dm^{-3} hydrochloric acid.

i)

Calculate the number of moles of copper(II) oxide and hydrochloric acid.

[2]

ii)

Explain which chemical is the limiting reagent.

[1]

[3 marks]

Question 4c

22.43 g of copper(II) hydroxide reacts with hydrochloric acid to form 23.63 g of copper(II) chloride.

Complete Table 4.2 to calculate the percentage yield of this reaction.

Table 4.2

Relative formula mass of copper(II) hydroxide	....
Moles of copper(II) hydroxide	....
Relative formula mass of copper(II) chloride	....
Theoretical yield of copper(II) chloride	....
Percentage yield of copper(II) chloride	....

[3 marks]

Question 4d

22.23 g of copper(II) carbonate reacts with hydrochloric acid at room temperature and pressure. One of the products is a gas.

You should use your answer to part **(a)(ii)** to help answer this question.

Calculate the volume, in cm^3, of the gas produced.

[2 marks]

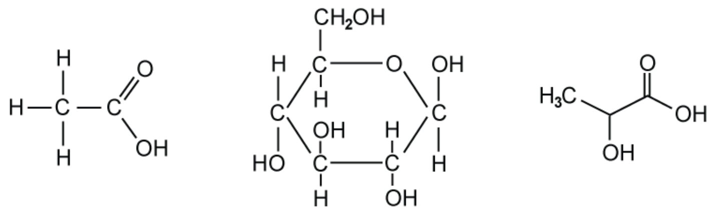
Question 5a

Explain the meaning of the terms empirical formula and molecular formula.

[2 marks]

Question 5b

Three organic compounds are shown in Fig. 5.1.



Acetic acid　　Glucose　　Lactic acid

Fig. 5.1

i)
Give the molecular formula of each compound.

Acetic acid ....

Glucose ....

Lactic acid ....

[3]

ii)
Give the empirical formula of all three compounds.

[1]

[4 marks]

Question 5c

An unknown hydrocarbon contains 85.7% carbon. The relative molecular mass of the unknown hydrocarbon is 56.0.

i)
Calculate the empirical formula of the unknown hydrocarbon. Show your working.

[3]

ii)
Determine the molecular formula of the unknown hydrocarbon. Show your working.

[2]

[5 marks]

Question 5d

Explain the difference between an anydrous salt and a hydrated salt.

[1 mark]

Question 5e

Gypsum is a salt that can contain water of crystallisation. In its anhydrous form, the chemical formula of gypsum is $CaSO_4$.
1.80 g of water is added to 6.81 g of gypsum to form the hydrated salt, $CaSO_4 \bullet xH_2O$.

Calculate the value of x in $CaSO_4 \bullet xH_2O$. Show your working.

[3 marks]

Model Answers: Easy

1a

The terms relative atomic mass and relative isotopic mass mean:

Relative atomic mass:

EITHER

- The mass of an atom; [1 mark]
- Relative / compared to 1/12 (the mass) of (an atom of) carbon-12
  OR
  On a scale in which a carbon-12 (atom / isotope) has (a mass of exactly) 12 (units); [1 mark]

OR

- The mass of one mole of atoms; [1 mark]
- Relative / compared to 1/12 (the mass) of 1 mole of carbon-12
  OR
  In which one mole of carbon-12 (atom / isotope) has a mass of (exactly) 12 g; [1 mark]

OR

- $\dfrac{\text{The average mass of one atom}}{{}^{1}\!/_{12}\text{ of the mass of one carbon} - 12\text{ atom}}$; [2 marks]

Relative isotopic mass:

EITHER

- The mass of an atom / isotope; [1 mark]]
- Relative / compared to 1/12 (the mass) of (an atom of) carbon-12
  OR
  On a scale in which a carbon-12 (atom / isotope) has (a mass of exactly) 12 (units); [1 mark]

OR

- The mass of one mole (of atoms) of an isotope; [1 mark]
- Relative / compared to 1/12 (the mass) of 1 mole of carbon-12
  OR
  In which one mole of carbon-12 (atom / isotope) has a mass of (exactly) 12 g; [1 mark]

[Total: 4 marks]

- These are two definitions that you need to learn
  - It is unlikely that they will both appear in the same series of exams

1b

i) The symbols used for relative atomic mass and relative molecular mass are:

- Relative atomic mass = A_r
  AND
  Relative molecular mass = M_r; [1 mark]

ii) To calculate the relative molecular mass of a compound:

- Add the (relative) atomic masses / A_r of all the atoms in the compound; [1 mark]

iii) The difference between relative molecular mass and relative formula mass is that:

- Relative molecular mass is for molecular compounds
  AND
  Relative formula mass is for compounds that contain ions; [1 mark]

[Total: 3 marks]

- You may see questions using the symbols A_r and M_r instead of the terms relative atomic mass and relative molecular mass
- You should be aware of these terms and how to calculate the M_r of a compound from your previous studies
- It is not absolutely necessary to know this difference but it can aid your understanding when using the terms
  - However, it can save you from losing a mark on a particularly strict mark scheme

1c

The formula unit of sodium chloride is:

- NaCl; [1 mark]

[Total: 1 mark]

- **Remember:** The formula unit is the simplest formula for a compound with a giant structure, for example:
  - The formula unit for silica / silicon dioxide is SiO_2
  - The formula unit for sodium chloride is NaCl
  - The formula unit of diamond is C

1d

The completed table is:

Chemical	Formula	Relative molecular / formula mass
Magnesium oxide	MgO	40.3; [1 mark]
Sulfuric acid	H_2SO_4	98.1; [1 mark]
Ammonium phosphate	$(NH_4)_3PO_4$	149.0; [1 mark]
Copper(II) sulfate pentahydrate	$CuSO_4•5H_2O$	249.6; [1 mark]

[Total: 4 marks]

- The most challenging part of these calculations is making sure that you are using the correct number of each atom
  - Also, use the values from your periodic table and don't guess / remember them from your previous studies because the relative atomic mass values might be more precise now
- MgO contains one magnesium atom and one oxygen atom
  - 24.3 + 16.0 = 40.3
- H_2SO_4 contains 2 hydrogen atoms, one sulfur atom and four oxygen atoms
  - (1.0 x 2) + 32.1 + (16.0 x 4) = 98.1
- $(NH_4)_3PO_4$

  - The ammonium portion contains one nitrogen and 4 hydrogen atoms
  - But, there are three of these
    - ((14 + (1.0 x 4)) x 3) = 54.0
  - The phosphate portion contains one phosphorous atom and 4 oxygen atoms
    - 31.0 + (16.0 x 4) = 95.0
  - These combine to give the overall relative formula mass of $(NH_4)_3PO_4$
    - 54.0 + 95.0 = 149.0
- $CuSO_4{\bullet}5H_2O$ is a hydrated copper(II) sulfate salt that contains 5 moles of water (water of crystallisation)
  - The $CuSO_4$ portion contains one copper atom, one sulfur atom and 4 oxygen atoms
    - 63.5 + 32.1 + (16.0 x 4) = 159.6
  - The $5H_2O$ portion contains 5 water molecules
    - Each water molecule contains 2 hydrogen atoms and one oxygen atom
      - (1.0 x 2) + 16.0 = 18.0
    - So, 5 water molecules = 18.0 x 5 = 90.0
  - These combine to give the overall relative formula mass of $CuSO_4{\bullet}5H_2O$
    - 159.6 + 90.0 = 249.6

2a

The number of particles in one mole of a chemical is:

- 6.02×10^{23}
  OR
  6.022×10^{23}; [1 mark]

[Total: 1 mark]

- The number of particles in one mole of a chemical is Avogadro's constant, *L*
- There are various values for Avogadro's constant, *L*, depending on the number of significant figures that an answer is written to
  - The value that is written in this specification is 6.02×10^{23}, which tells you that there are 6.02×10^{23} particles in one mole of any chemical
- It is more likely that you will be given this value in questions but it can help you answer questions quicker if you know it

2b

i) The number and type of atoms present in a molecule of water are:

- 3 / three atoms; [1 mark]
- 2 / two hydrogen atoms
  AND
  1 / one oxygen atom; [1 mark]

ii) The total number of particles and atoms present in one mole of water is:

- Particles = 6.02 x 10^{23}
  AND
  Atoms = 1.806 x 10^{24}; [1 mark]

[Total: 3 marks]

- From your previous studies, you should be able to describe the composition of a water molecule in terms of the number and type of atoms present
  - H_2O - the sub-script number 2 belongs to the hydrogen atoms, telling you that there are two hydrogen atoms
  - There is no sub-script number after the oxygen, which means that there is only one oxygen atom
- Avogadro's constant tells you the number of particles present in one mole of a chemical
  - The word particles can mean atoms, molecules, ions or electrons
- Since you are not given any more information about the particles, you have to assume that the word particles is being used to mean molecules
  - Therefore, there are 6.02 x 10^{23} particles / molecules in one mole of water
- Since water contains three atoms, the total number of atoms in one mole of water is 3 x Avogadro's constant

AND

Atoms = 1.806×10^{24}; [1 mark]

[Total: 3 marks]

- From your previous studies, you should be able to describe the composition of a water molecule in terms of the number and type of atoms present
  - H_2O - the sub-script number 2 belongs to the hydrogen atoms, telling you that there are two hydrogen atoms
  - There is no sub-script number after the oxygen, which means that there is only one oxygen atom
- Avogadro's constant tells you the number of particles present in one mole of a chemical
  - The word particles can mean atoms, molecules, ions or electrons
- Since you are not given any more information about the particles, you have to assume that the word particles is being used to mean molecules
  - Therefore, there are 6.02×10^{23} particles / molecules in one mole of water
- Since water contains three atoms, the total number of atoms in one mole of water is 3 x Avogadro's constant
  - $3 \times 6.02 \times 10^{23} = 1.806 \times 10^{24}$ atoms in one mole of water

2c

i) The number and type of ions present in one mole of sodium carbonate are:

- 2 sodium / Na^+ ions; [1 mark]
- 1 carbonate / CO_3^{2-} ion; [1 mark]

ii) The number of metal ions present in **two** moles of sodium carbonate is:

- Number of metal / sodium ions in **one** mole = 1.204×10^{24}; [1 mark]
- Number of metal / sodium ions in **two** moles = 2.408×10^{24}; [1 mark]

[Total: 4 marks]

- Sodium carbonate, Na_2CO_3, contains the sodium ion and the carbonate ion
  - The sodium ion has a formula of Na^+ since it is a group 1 metal and loses one electron to form an ion
  - Since there is a sub-script 2 after the Na symbol, there are 2 sodium / Na^+ ions
  - This means that the carbonate ion must have a 2– charge to give a neutral

compound overall
    - Therefore, the carbonate ion is CO_3^{2-}
        - This is an ion that you can be expected to know
- You are asked for the number of metal ions present in one mole of sodium carbonate
    - The metal is sodium
    - This means that you need to double Avogadro's constant to get the number of sodium / metal ions in one mole of sodium carbonate
    - **Remember:** Avogadro's constant tells you the number of particles present in one mole of a chemical
    - You then need to double your answer again to get the number of sodium / metal ions in two moles of sodium carbonate
- Overall, the calculation is $6.02 \times 10^{23} \times 4 = 2.408 \times 10^{24}$ sodium / metal ions in two moles of sodium carbonate

2d

i) The conditions when one mole of a gas occupies a volume of 24.0 dm^3 are:

- 20 ^{o}C
  OR
  Room temperature; [1 mark]
- 1 atmosphere
  OR
  Room pressure; [1 mark]

ii) The completed Table 2.1 is:

Gas	Number of moles	Volume of gas (dm^3)	Number of molecules present
Nitrogen	2.0	48.0	1.204×10^{24}; [1 mark]
Sulfur dioxide	0.075; [1 mark]	1.8	4.515×10^{22}; [1 mark]
Carbon monoxide	1.5; [1 mark]	36.0; [1 mark]	9.03×10^{23}

[Total: 7 marks]

- You need to know that any gas occupies 24.0 dm^3 at room temperature and pressure
  - The values for room temperature and pressure are 20 ^oC and 1 atmosphere respectively
- The information for nitrogen in Table 2.1 shows you that the gases are at room temperature and pressure because 2.0 moles of nitrogen has a volume of 48.0 dm^3
- Nitrogen
  - Two moles of nitrogen will contain 2 x 6.02 x 10^{23} = 1.204 x 10^{24} molecules
- Sulfur dioxide
  - 24.0 dm^3 of sulfur dioxide is one mole
  - Therefore, 1.8 dm^3 is $\frac{1.8}{24}$ = 0.075 moles of sulfur dioxide
  - 0.075 moles of sulfur dioxide will contain 0.075 x 6.02 x 10^{23} = 4.515 x 10^{22} molecules
- Carbon monoxide
  - 6.02 x 10^{23} molecules is one mole
  - Therefore, 9.06 x 10^{23} is $\frac{9.06 \times 10^{23}}{6.02 \times 10^{23}}$ = 1.5 moles of carbon monoxide
  - 1.5 moles of carbon monoxide will have a volume of 1.5 x 24.0 = 36.0 dm^3

3a

The completed Table 3.1 is:

Ion	Formula
Nitrate	NO_3^-
Carbonate	CO_3^{2-}
Sulfate	$\mathbf{SO_4^{2-}}$
Hydroxide	OH^-
Ammonium	NH_4^+
Zinc(II)	$\mathbf{Zn^{2+}}$
Silver(I)	Ag^+
Hydrogen carbonate	HCO_3^-
Phosphate	$\mathbf{PO_4^{3-}}$

- All 8 ions and formulae correct; [3 marks]
- 4 - 7 ions and formulae correct; [2 marks]
- 1 - 3 ions and formulae correct; [1 mark]

[Total: 3 marks]

- All of the ions and formulae in this question are the ones that you are expected to know for this course
- You also need to be able to use the periodic table to write the formula of ions formed by elements, for example:
  - Oxygen forms the oxide ion, O^{2-}
  - Halogens typically form the halide ion, X^-
  - Group 1 metals form group 1 metal ions, M^+
- You should also be able to use roman numerals to identify ions as shown by zinc(II) and Ag^+ in this question

3b

The completed Table 3.2 is:

Ionic compound	Formula
Sodium chloride	NaCl
Potassium oxide	K_2O; [1 mark]
Aluminium carbonate; [1 mark]	$Al_2(CO_3)_3$
Copper(II) nitrate	$Cu(NO_3)_2$; [1 mark]

[Total: 3 marks]

- Potassium oxide
  - Potassium is in Group 1 so its ion has a charge of 1+, K^+
  - Oxygen is in Group 16 so the oxide ion has a charge of 2–, O^{2-}
  - Two potassium ions are needed for each oxide ion
  - So, the formula is K_2O
- $Al_2(CO_3)_3$
  - From the chemical formula, this contains the aluminium ion (Al^{3+}) and the carbonate ion (CO_3^{2-})
  - Therefore, the name of the ionic compound is aluminium carbonate
- Copper(II) nitrate
  - The roman numeral attached to copper shows that the copper ion has a charge of 2+, Cu^{2+}

- The nitrate ion has a charge of 1–, NO_3^-
- Two nitrate ions are needed for each copper(II) ion
- **Remember:** The nitrate ion will have to be inside brackets
- So, the formula is $Cu(NO_3)_2$

3c

i) The equation for the reaction of silver nitrate with hydrochloric acid is:

$AgNO_3$ (aq) + HCl (aq) → AgCl (s) + HNO_3 (aq)

- All chemical formulae correct; [1 mark]
- All state symbols correct; [1 mark]

ii) Nitrate ions and hydrogens do not appear in the ionic equation because:

- They are spectator ions
  OR
  They are not (actively) involved in the reaction / formation of silver chloride
  OR
  An ionic equation only shows the ions / particles that take in the reaction; [1 mark]

[Total: 3 marks]

- You can build the word equation from the information in the question
  - Silver nitrate + hydrochloric acid → silver chloride + nitric acid
- The silver ion has a 1+ charge, Ag^+
- The nitrate ion has a 1– charge, NO_3^-
  - So, the formula of silver nitrate is $AgNO_3$
- You are given the formula of silver chloride and are expected to know the formulae of hydrochloric and nitric acid

  - **Remember:** The nitrate ion will have to be inside brackets
  - So, the formula is $Cu(NO_3)_2$

3c

i) The equation for the reaction of silver nitrate with hydrochloric acid is:

$AgNO_3$ (aq) + HCl (aq) → AgCl (s) + HNO_3 (aq)

- All chemical formulae correct; [1 mark]
- All state symbols correct; [1 mark]

ii) Nitrate ions and hydrogens do not appear in the ionic equation because:

- They are spectator ions
  OR
  They are not (actively) involved in the reaction / formation of silver chloride
  OR
  An ionic equation only shows the ions / particles that take in the reaction; [1 mark]

[Total: 3 marks]

- You can build the word equation from the information in the question
  - Silver nitrate + hydrochloric acid → silver chloride + nitric acid
- The silver ion has a 1+ charge, Ag^+
- The nitrate ion has a 1– charge, NO_3^-
  - So, the formula of silver nitrate is $AgNO_3$
- You are given the formula of silver chloride and are expected to know the formulae of hydrochloric and nitric acid
- Replacing the words for the chemical formulae gives:
  - $AgNO_3$ + HCl → AgCl + HNO_3
- The question tells you that the silver nitrate is aqueous / (aq) and that the silver chloride is a precipitate, which means that it is solid / (s)
- You are expected to know that hydrochloric and nitric acid are aqueous / (aq)
- Adding the state symbols gives:
  - $AgNO_3$ (aq) + HCl (aq) → AgCl (s) + HNO_3 (aq)
- **Remember:** Ionic equations only show the ions that are reacting to form the desired product
  - Any other ions are not included - they are known as spectator ions

- It is very unlikely that a question would ask you why certain ions do not appear in an ionic equation but it is important that you know spectator ions are not included

3d

i) The chemical equation for the reaction of barium chloride with sulfuric acid is:

$BaCl_2$ (aq) + H_2SO_4 (aq) → $BaSO_4$ (s) + 2HCl (aq)

- All chemical formulae correct; [1 mark]
- All state symbols correct
  AND
  Correct balancing; [1 mark]

ii) The ionic equation for the formation of barium sulfate is:

- Ba^{2+} (aq) + SO_4^{2-} (aq) → $BaSO_4$ (s); [1 mark]

[Total: 3 marks]

- You can build the word equation from the information in the question
  - Barium chloride + sulfuric acid → barium sulfate + hydrochloric acid
- The barium ion has a 2+ charge, Ba^{2+}
- The chloride ion has a 1– charge, Cl^-
  - So, the formula of barium chloride is $BaCl_2$
- The sulfate ion has a 2– charge, SO_4^{2-}
  - So, the formula of barium sulfate is $BaSO_4$
- You are expected to know the formulae of sulfuric and hydrochloric acid
- Replacing the words and adding the state symbols gives:
  - $BaCl_2$ (aq) + H_2SO_4 (aq) → $BaSO_4$ (s) + HCl (aq)
- The equation is unbalanced as there are two chlorines and two hydrogens on the left-hand side and only one of each on the right-hand side
  - This can be resolved by doubling the amount of HCl produced
  - $BaCl_2$ (aq) + H_2SO_4 (aq) → $BaSO_4$ (s) + **2**HCl (aq)
- **Remember:** Ionic equations only show the ions that are reacting to form the desired product
  - Any other ions are not included - they are known as spectator ions
  - The reactant ions are the barium ion and the sulfate ion, which form the barium

sulfate precipitate
    - Ba^{2+} (aq) + SO_4^{2-} (aq) → $BaSO_4$ (s)
- Examiners should indicate if they want you to include state symbols in chemical / ionic equations but you should always include them so you do not potentially lose a mark

4a

i) The reaction of copper with hydrochloric acid is not used to form copper(II) chloride because:

- Copper is unreactive
  OR
  Copper is lower in the reactivity series than hydrogen; [1 mark]

ii) The chemical equations for the formation of copper(II) chloride using hydrochloric acid and the copper(II) compounds are:

Copper(II) oxide:

- $CuO + 2HCl \rightarrow CuCl_2 + H_2O$; [1 mark]

Copper(II) hydroxide:

- $Cu(OH)_2 + 2HCl \rightarrow CuCl_2 + 2H_2O$; [1 mark]

Copper(II) carbonate:

- $CuCO_3 + 2HCl \rightarrow CuCl_2 + H_2O + CO_2$; [1 mark]

[Total: 4 marks]

- From your previous studies, you should be able to explain why copper is not reacted with hydrochloric acid to form copper(II) chloride
    - Copper metal is relatively unreactive is a basic answer
    - Copper is below hydrogen in the reactivity series is a stronger answer
- You should also know the reactions of metal oxides / hydroxides / carbonates with acid from your previous studies
    - Copper(II) has a 2+ charge so its oxides / hydroxides / carbonates will react with acid in a similar fashion to the group 2 metals
- **Remember:** Metal oxide + acid → metal salt + water

- The unbalanced equation for copper(II) oxide with HCl is:
  - $CuO + HCl \rightarrow CuCl_2 + H_2O$
- There are 2 hydrogen and 2 chlorine atoms on the right-hand side but only 1 hydrogen and 1 chlorine atom on the left-hand side, so you double the HCl
  - $CuO + \mathbf{2}HCl \rightarrow CuCl_2 + H_2O$

- **Remember:** Metal hydroxide + acid → metal salt + water
  - The unbalanced equation for copper(II) hydroxide with HCl is:
    - $Cu(OH)_2 + HCl \rightarrow CuCl_2 + H_2O$
  - There are 2 chlorine atoms on the right-hand side and only1 chlorine atom on the left-hand side, so you double the HCl
    - $Cu(OH)_2 + \mathbf{2}HCl \rightarrow CuCl_2 + H_2O$
  - There are 4 hydrogen and 2 oxygen atoms on the left-hand side and only 2 hydrogen atoms and 1 oxygen atom on the right-hand side, so you double the H_2O
    - $Cu(OH)_2 + \mathbf{2}HCl \rightarrow CuCl_2 + \mathbf{2}H_2O$
- **Remember:** Metal carbonate + acid → metal salt + water + carbon dioxide
  - The unbalanced equation for copper(II) carbonate with HCl is:
    - $CuCO_3 + HCl \rightarrow CuCl_2 + H_2O + CO_2$
  - There are 2 hydrogen and 2 chlorine atoms on the right-hand side but only 1 hydrogen and 1 chlorine atom on the left-hand side, so you double the HCl
    - $CuCO_3 + \mathbf{2}HCl \rightarrow CuCl_2 + H_2O + CO_2$

4b

i) The number of moles of copper(II) oxide and hydrochloric acid are:

- Formula mass of copper(II) oxide = 79.5
  AND
  Moles of copper(II) oxide = 0.15; [1 mark]
- Moles of hydrochloric acid = 0.1; [1 mark]

ii) The limiting reagent is:

- Hydrochloric acid
  AND
  (Because,) there are less / fewest moles of hydrochloric acid; [1 mark]

[Total: 3 marks]

- This question requires you to use both mole equations:
  - Moles = $\frac{\text{mass}}{M_r}$
  - Moles = concentration x volume
    - **Remember:** The volume is in dm^3
- In this reaction:
  - The copper(II) oxide is in excess because it has the most moles
  - Therefore, the hydrochloric acid is the limiting reagent because it has the least moles

4c

The values to complete Table 4.2 are:

- Relative formula mass of copper(II) hydroxide = 97.5
  AND
  Moles of copper(II) hydroxide = 0.230; [1 mark]
- Relative formula mass of copper(II) chloride = 134.5
  AND
  Theoretical yield of copper(II) hydroxide = 30.94; [1 mark]
- Percentage yield of copper(II) chloride = 76.4%; [1 mark]

[Total: 3 marks]

- Table 4.2 guides you through the steps required to calculate the percentage yield for the reaction
  1. Relative formula mass of copper(II) hydroxide
     - $Cu(OH)_2 = 63.5 + ((16.0 + 1.0) \times 2) = 97.5$
  2. Moles of copper(II) hydroxide
     - Moles = $\frac{\text{mass}}{M_r} = \frac{22.43}{97.5} = 0.230$

4b

i) The number of moles of copper(II) oxide and hydrochloric acid are:

- Formula mass of copper(II) oxide = 79.5
  AND
  Moles of copper(II) oxide = 0.15; [1 mark]
- Moles of hydrochloric acid = 0.1; [1 mark]

ii) The limiting reagent is:

- Hydrochloric acid
  AND
  (Because,) there are less / fewest moles of hydrochloric acid; [1 mark]

[Total: 3 marks]

- This question requires you to use both mole equations:
  - Moles = $\frac{\text{mass}}{M_r}$
  - Moles = concentration x volume
    - **Remember:** The volume is in dm^3
- In this reaction:
  - The copper(II) oxide is in excess because it has the most moles
  - Therefore, the hydrochloric acid is the limiting reagent because it has the least moles

4c

The values to complete Table 4.2 are:

- Relative formula mass of copper(II) hydroxide = 97.5
  AND
  Moles of copper(II) hydroxide = 0.230; [1 mark]
- Relative formula mass of copper(II) chloride = 134.5
  AND
  Theoretical yield of copper(II) hydroxide = 30.94; [1 mark]
- Percentage yield of copper(II) chloride = 76.4%; [1 mark]

[Total: 3 marks]

- Table 4.2 guides you through the steps required to calculate the percentage yield for the reaction
  1. Relative formula mass of copper(II) hydroxide
     - $Cu(OH)_2 = 63.5 + ((16.0 + 1.0) \times 2) = 97.5$
  2. Moles of copper(II) hydroxide
     - Moles $= \frac{\text{mass}}{M_r} = \frac{22.43}{97.5} = 0.230$
  3. Relative formula mass of copper(II) chloride
     - $CuCl_2 = 63.5 + (35.5 \times 2) = 134.5$
  4. Theoretical yield of copper(II) chloride
     - Mass = moles x $M_r = 0.230 \times 134.5 = 30.94$
  5. Percentage yield of copper(II) chloride
     - Percentage yield = $= \frac{\text{actual yield}}{\text{theoretical yield}} \times 100 = \frac{23.63}{30.94} \times 100 = 76.4\%$

4d

To calculate the volume, in cm^3, of the gas produced:

- Relative formula mass of copper(II) carbonate = 123.5
  AND
  Moles of copper(II) carbonate = 0.18; [1 mark]
- Volume of carbon dioxide = 4320 (cm^3); [1 mark]

[Total: 2 marks]

- To calculate the volume, in cm^3, of the gas produced:
  1. Calculate the relative formula mass of copper(II) carbonate
     - $CuCO_3 = 63.5 + 12.0 + (16.0 \times 3) = 123.5$
  2. Calculate the moles of copper(II) carbonate
     - Moles $= \frac{\text{mass}}{M_r} = \frac{22.23}{123.5} = 0.18$
  3. From your equation in part (a) (ii), one mole of copper(II) carbonate reacts to form one mole of carbon dioxide produced
     - Therefore, 0.18 moles of carbon dioxide are produced
  4. Calculate the volume of carbon dioxide produced
     - **Remember:** One mole of any gas at room temperature and pressure occupies 24 dm^3 or 24000 cm^3
     - Volume = moles x 24000 = 0.18 x 24000 = 4320 cm^3

5a

Empirical formula and molecular formula mean:

- Empirical formula = The simplest whole number ratio of the elements present in one molecule / formula unit of a compound; [1 mark]
- Molecular formula = The actual ratio of the elements present in one molecule / formula unit of a compound; [1 mark]

[Total: 2 marks]

- You need to know these definitions
- You should also be able to go from empirical to molecular formula and vice versa
  - This may be achieved by looking at structure diagrams or calculation

5b

i) The molecular formula of each compound is:

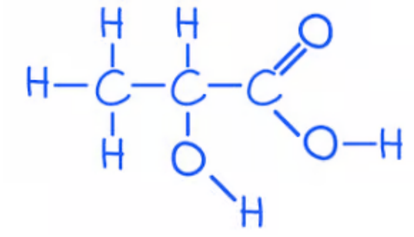
- Acetic acid = $C_2H_4O_2$; [1 mark]
- Glucose = $C_6H_{12}O_6$; [1 mark]
- Lactic acid = $C_3H_6O_3$; [1 mark]

ii) The empirical formula of all three compounds is:

- CH_2O; [1 mark]

[Total: 4 marks]

- For the molecular formula of acetic acid and glucose, you can count up the number of carbon, hydrogen and oxygen atoms
  - Acetic acid contains 2 carbons, 4 hydrogens and 2 oxygens giving the molecular formula $C_2H_4O_2$
  - Glucose contains 6 carbons, 12 hydrogens and 6 oxygens giving the molecular formula $C_6H_{12}O_6$
- **Careful:** The structure shown for lactic acid is not a fully displayed structure - the skeletal carbons are not shown and there is also a "missing" hydrogen



- Lactic acid contains 3 carbons, 6 hydrogens and 3 oxygens giving the molecular formula $C_3H_6O_3$
- The empirical formula / simplest whole-number ratio of atoms for all three compounds is CH_2O
  - They all have equal numbers of carbon and oxygen atoms and twice as many hydrogen atoms

5c

i) To calculate the empirical formula of the unknown hydrocarbon:

- Moles of carbon = 7.142
  AND
  Moles of hydrogen = 14.3; [1 mark]
- Ratio of carbon to hydrogen (divide by smallest)
  Carbon = $\frac{7.142}{7.142}$ = 1
  AND
  Hydrogen = $\frac{14.3}{7.142}$ = 2.00; [1 mark]
- Empirical formula = CH_2; [1 mark]

ii) To determine the molecular formula of the unknown hydrocarbon:

- Mass of CH_2 = 14.0
  AND
  $\frac{56.0}{14.0}$ = 4; [1 mark]
- Molecular formula = C_4H_8; [1 mark]

[Total: 5 marks]

- Empirical formula can be consistently answered using the following template for the calculation

-

Elements	Carbon	Hydrogen
Value (in g or %)	85.7	14.3
Atomic mass	12.0	1.0
Divide (moles)	$\frac{85.7}{12.0} = 7.142$	$\frac{14.3}{1.0} = 14.3$
Ratio (divide by smallest)	$\frac{7.142}{7.142} = 1$	$\frac{14.3}{7.142} = 2.00$
Answer	C	H_2

- To calculate the molecular mass, you need to find the mass of the empirical formula
  - CH_2 = 12.0 + (1.0 x 2) = 14.0
- Then divide the relative molecular mass by the mass of the empirical formula
  - $\frac{56.0}{14.0} = 4$
  - This tells you that there are 4 CH_2 in the molecular formula
  - 4 x CH_2 = C_4H_8

5d

The difference between an anydrous salt and a hydrated salt is:

- An anhydrous salt contains no water (of crystallisation)
  OR
  A hydrated salt contains water (of crystallisation); [1 mark]

[Total: 1 mark]

- The *hydr* portions of anhydrous and hydrated should guide you towards the difference being about water

- Empirical formula can be consistently answered using the following template for the calculation
-

Elements	Carbon	Hydrogen
Value (in g or %)	85.7	14.3
Atomic mass	12.0	1.0
Divide (moles)	$\frac{85.7}{12.0} = 7.142$	$\frac{14.3}{1.0} = 14.3$
Ratio (divide by smallest)	$\frac{7.142}{7.142} = 1$	$\frac{14.3}{7.142} = 2.00$
Answer	C	H_2

- To calculate the molecular mass, you need to find the mass of the empirical formula
  - CH_2 = 12.0 + (1.0 x 2) = 14.0
- Then divide the relative molecular mass by the mass of the empirical formula
  - $\frac{56.0}{14.0} = 4$
  - This tells you that there are 4 CH_2 in the molecular formula
  - 4 x CH_2 = C_4H_8

5d

The difference between an anydrous salt and a hydrated salt is:

- An anhydrous salt contains no water (of crystallisation)
  OR
  A hydrated salt contains water (of crystallisation); [1 mark]

[Total: 1 mark]

- The *hydr* portions of anhydrous and hydrated should guide you towards the difference being about water

5e

To calculate the value of x in $CaSO_4 \bullet xH_2O$:

- Mass of $CaSO_4$ = 136.2
  AND

Moles of $CaSO_4$ = 0.05; [1 mark]

- Mass of H_2O = 18.0
  AND
  Moles of H_2O = 0.1; [1 mark]
- Ratio of $CaSO_4$ to H_2O = 1 : 2
  AND
  The value of x = 2 **OR** the formula is $CaSO_4{\bullet}2H_2O$; [1 mark]

[Total: 3 marks]

- Water of crystallisation calculations are a variation on empirical formula calculations
- They can be consistently solved using a variation of the template seen in part (c)
- You need to calculate the relative formula mass of $CaSO_4$ and H_2O
  - $CaSO_4$ = 40.1 + 32.1 + (16.0 x 4) = 136.2
  - H_2O = (1.0 x 2) + 16.0 = 18.0
-

~~Elements~~	**Compounds**	$CaSO_4$	H_2O
Value		6.81 g	1.80 g
~~Atomic mass~~	**Molecular mass**	136.2	18.0
Divide (moles)		$\frac{6.81}{136.2} = 0.05$	$\frac{1.80}{18.0} = 0.1$
Ratio (divide by smallest)		$\frac{0.05}{0.05} = 1$	$\frac{0.1}{0.05} = 2$
Answer		x=2	

- The ratio step gives you the ratio of salt to water or 1 : x
  - In this case, the value of x is 2
  - This can also be shown in your answer by writing the actual formula of the hydrated salt, $CaSO_4{\bullet}2H_2O$